

M3: Complete Implementation

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Abstract—This project milestone explores patterns in movie ratings and user behavior using clustering, classification, and temporal analysis techniques. Movie clusters were identified using K-Means clustering based on rating, popularity, and genre features, revealing groups defined primarily by audience engagement rather than perceived quality. User archetypes were discovered through clustering on behavioral features such as rating frequency, consistency, and tagging activity. Decision trees were used to interpret and validate both movie and user clusters, highlighting the most influential features. Temporal analysis further revealed lifecycle patterns in user activity, including initial bursts of engagement followed by steep decline. These findings provide insights into how users interact with movies and how behavior evolves over time.

I. INTRODUCTION

This project analyzes the MovieLens dataset to uncover meaningful patterns in both movies and user behavior. The goal is to move beyond simple statistics and identify natural groupings and behavioral trends using data mining techniques.

Discovery Questions:

- Q1: What natural groups of movies exist based on user ratings, genres, and tags?
- Q2: How do user engagement patterns (rating volume and tagging) relate to rating behavior, and how do these factors form distinct user archetypes?
- Q3: How do temporal patterns in rating activity (burst behavior, rating recency, and consistency over time) differ between user archetypes, and what anomalous rating patterns emerge?

II. IMPLEMENTATION STATUS

The core clustering techniques from M2 are fully implemented and operational. For movies, K-means clustering with principal component analysis preprocessing identified four distinct clusters, with the optimal number of clusters determined through silhouette score analysis and the elbow method applied to within-cluster sum of squares. For users, similar K-means analysis identified two distinct archetypes, determined through silhouette score analysis, confirming reasonable separation between the groups. Both clustering results are saved to CSV files and have been thoroughly characterized in terms of their defining features.

For M3 expansion, three additional techniques have been successfully implemented. First, decision tree classification was applied to both movie and user clusters, treating cluster assignments as class labels to validate that the discovered

groups represent real patterns rather than algorithmic artifacts. For movies, this involved building a decision tree excluding the potentially problematic `metadata_richness` feature to avoid circular logic, using only average rating, popularity log, and genre indicators as predictors. For users, decision trees were built using behavioral features only (average rating, rating standard deviation, total ratings, tags given, and maximum daily ratings) to directly address the discovery question about engagement patterns. Second, a comprehensive temporal analysis was implemented to examine how rating behavior evolves over time. This involved session detection using two-hour gaps to identify distinct rating sessions, calculation of session lengths, analysis of first versus later session behavior, rating recency measured as years after movie release, and user lifespan calculated as days between first and last rating. Third, anomaly detection using Local Outlier Factor was implemented to identify users with unusual engagement patterns. This analysis was performed to answer the last part of question 3 about anomalous rating patterns.

All data pipelines and analyses execute successfully and produce consistent, interpretable results across multiple validation methods.

III. ANALYTICAL CHOICES

Several analytical choices were explored and evaluated throughout this project, with final decisions driven by the need to produce interpretable, valid results that directly address the discovery questions.

A. Movie Clustering

For movie (K-Means) clustering, the initial approach included `metadata_richness` as a feature alongside average rating, popularity log, and genre indicators. This feature represents the total count of tags, descriptions, and metadata associated with each movie. While seemingly informative, a decision tree analysis revealed that `metadata_richness` dominated the clustering with 100% feature importance, meaning the clusters were essentially grouping movies by how many tags they had. This raised serious concerns about circular logic, as more popular movies naturally attract more tags and metadata due to higher user engagement. To resolve this, two approaches were explored: re-clustering without `metadata_richness` entirely, versus keeping the original clusters but validating them through a decision tree that excluded the problematic feature. The re-clustering approach produced non-sensical results where a cluster called Popular Hits had only

50 average ratings per movie, and a cluster called All-Time Classics had only 37 average ratings. This demonstrated that `metadata_richness`, while problematic as a clustering feature, was actually capturing meaningful engagement patterns. The chosen approach was therefore to keep the original clusters and use a decision tree without `metadata_richness` to validate that these clusters represent real patterns driven by genuine audience engagement metrics. This decision allows the report to present clean, interpretable clusters while transparently addressing the metadata concern through rigorous validation.

B. User Clustering

For user (K-Means) clustering, a similar exploration revealed that genre preferences were dominating the clustering when included alongside behavioral features. A decision tree using all features showed Adventure genre at 88.2% importance, meaning the user archetypes were being defined by what users watched rather than how they engaged. Since the discovery question specifically asks about engagement patterns, rating volume and tagging behavior, this approach did not answer the intended question. The alternative explored was to re-cluster users using only behavioral features, but this would discard the meaningful genre patterns discovered. The chosen approach was to keep the original user clusters but build a separate decision tree using only behavioral features to validate that engagement patterns form distinct archetypes. This yields two complementary insights: the original clusters show that users have different genre preferences (Adventure seekers vs Drama enthusiasts), while the behavioral decision tree shows that these same groups also differ significantly in how they rate and tag. This dual analysis provides a richer understanding of user archetypes than either approach alone.

C. Temporal Analysis

For temporal analysis, several session gap definitions were explored, including one, two, and four-hour thresholds. The two-hour gap was selected because it represents a reasonable boundary between distinct rating sessions while being robust to brief interruptions. For rating recency, the calculation of years after release using the datetime data and extracted release years from movie titles required careful handling of missing values and outlier detection. The decision to analyze recency by user type allows direct comparison between the two archetypes, addressing the part of Q3 about how temporal patterns differ.

D. Decision Tree Depth

The decision tree depth was another considered choice. Shallower trees (depth 3-4) were selected over deeper trees (depth 8-10) to prioritize pattern discovery and interpretability over predictive accuracy. A depth of 4 produces rules that are human-readable and informative while still achieving 95.5% accuracy for movies, confirming that the trade-off between interpretability and accuracy was appropriate.

E. Dimensionality Reduction with PCA

To prepare the high-dimensional feature space for clustering, Principal Component Analysis was applied to both movie and user datasets. PCA served two critical purposes by reducing noise and multicollinearity among correlated features, and enabling visualization of the cluster structure in lower-dimensional space.

For movies, the feature space consisted of 24 dimensions including average rating, popularity log, and 20 genre indicators. PCA revealed that the first principal component alone explained 99.99% of the total variance, with the first two components capturing nearly 100%. This extraordinary concentration of variance indicates that movie clusters are driven by a single dominant dimension; audience engagement, which aligns with later validation findings.

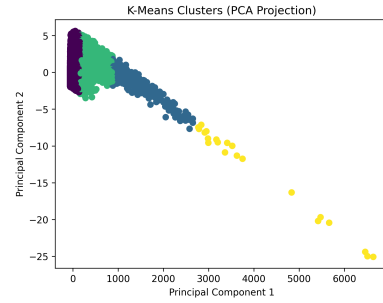


Fig. 1. Movie Clusters

For users, the feature space included behavioral metrics (rating volume, consistency, tagging activity) alongside genre preference vectors. PCA was applied to reduce dimensionality for visualization, with the first two principal components explaining 33.8% of the total variance (17.5% from PC1 and 16.4% from PC2). While this represents a moderate amount of explained variance, the PCA scatter plot demonstrates clear separation between the two user archetypes, confirming that the underlying behavioral patterns are captured in the reduced space.

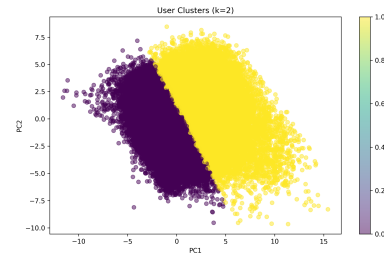


Fig. 2. User Clusters

NOTE: The two PCA scatter plots shown above demonstrate that after dimensionality reduction, both movie and user clusters exhibit reasonable separation in the transformed space, validating that the clustering algorithm operates on meaningful latent dimensions rather than noisy, high-dimensional signals.

F. Anomaly Detection

Local Outlier Factor (LOF) was used with $n_neighbors=20$ and $contamination='auto'$. The $n_neighbors$ parameter was set to 20 to balance local sensitivity with statistical stability, following standard practice for datasets of this scale. The $contamination$ parameter was set to 'auto' to let the algorithm determine the outlier threshold based on the natural distribution of outlier scores, rather than forcing a predetermined percentage. This approach ensures that the identified outliers represent genuinely anomalous behavior patterns as determined by the data itself.

IV. RESULTS

A. Natural Groups in Movies

The clustering analysis identified four distinct groups of movies forming a hierarchy of audience engagement.

Cluster	Count	Percentage	Avg Rating	Avg Rating Count	Top Genres
Long Tail	29,982	93.8%	3.14	426	Drama, Comedy, Thriller
Mid-Tier Successes	1,700	5.3%	3.47	6,915	Drama, Thriller, Comedy, Action
Popular Hits	258	0.8%	3.83	23,742	Action, Sci-Fi, Adventure
All-Time Classics	21	0.07%	4.13	56,145	Drama, Thriller, Sci-Fi

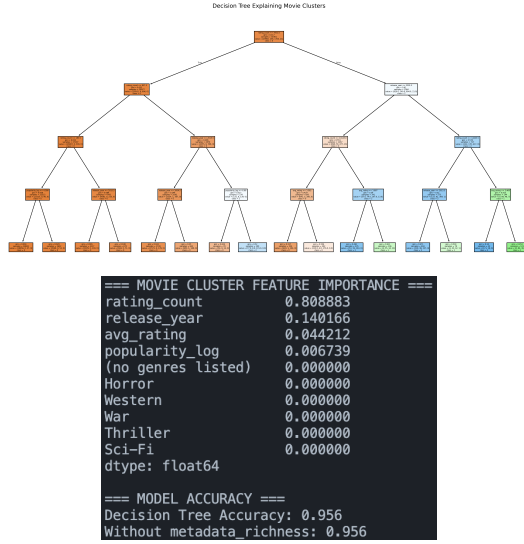


Fig. 3. Movie Decision Tree

Validation: Silhouette score for $k=4$ was 0.879, with the elbow method confirming $k=4$ as optimal. The decision tree validation (excluding $metadata_richness$) achieved 95.5% accuracy. Feature importance shows $rating_count$ (80.9%) dominates, followed by $release_year$ (14.0%) and avg_rating (4.4%), confirming that audience engagement drives cluster membership 18 times more than critical acclaim.

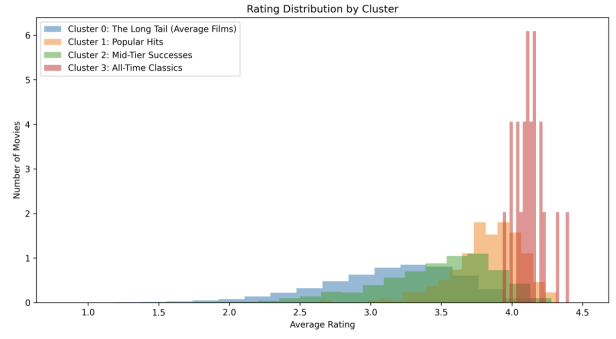


Fig. 4. Rating Distribution

Key Insight: Movie groups form a hierarchy of audience engagement. A movie becomes a “Popular Hit” because people watch it (Rating Count = 80.9%), not because it’s highly rated (Avg Rating = 4.4%). Classics are the rare exception, requiring both popularity and quality.

B. User Archetypes from Engagement Patterns

The clustering analysis identified two distinct user archetypes that differ in both genre preferences and rating behavior.

Archetype	Engaged Raters	Generous Raters
Percentage	65.4%	34.6%
Top Genres	Drama, Comedy, Romance	Action, Adventure, Drama
Avg Rating Count	175	129
Tags/User	12.1	5.8
Lifespan (days)	289	202
Avg Rating	3.66	3.79
Rating Decline	-0.22	-0.17

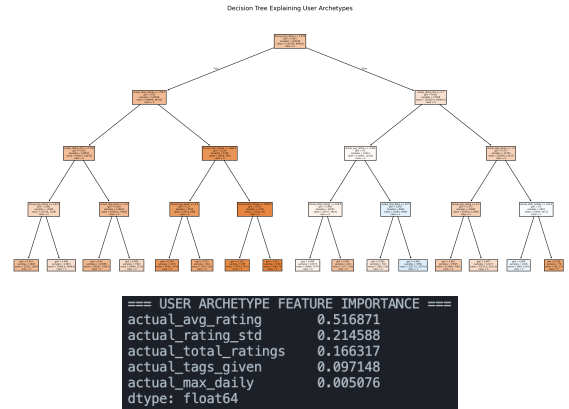


Fig. 5. User Decision Tree

Validation: The silhouette score for $k=2$ was 0.127, indicating modest but meaningful separation between the two archetypes. Higher k values produced lower silhouette scores, confirming that two clusters optimally capture the natural structure in user behavior. For users, the first two PCA components captured 33.8% of the variance, with PC1 explaining 17.5%

and PC2 explaining 16.4%. Although this moderate explained variance suggests user behavior is more complex, the PCA scatter plot still shows clear separation between the two archetypes.

The decision tree built using only behavioral features shows that average rating accounts for 51.7% of feature importance, rating standard deviation accounts for 21.5%, total ratings accounts for 16.6%, and tags given accounts for 9.7%. This indicates that how users rate, whether they give high-consistent ratings, matters more than what they watch or how much they rate.

A particularly noteworthy finding is the relatively low importance of tagging behavior. While Engaged Raters tag more frequently (12.1 vs 5.8 tags per user), tagging accounts for only 9.7% of the predictive power, substantially less than rating quality at 51.7%. This indicates that tagging is largely independent of other engagement metrics. A user can be highly engaged in rating without being an active tagger, and conversely, some moderate raters may tag extensively. This suggests rating and tagging represent distinct dimensions of user engagement: rating reflects consumption and evaluation, while tagging reflects a more participatory, community-oriented form of engagement.

C. Temporal Patterns and Anomalies Across User Archetypes

The temporal analysis reveals how the rating behavior evolves over time and differs between user archetypes.

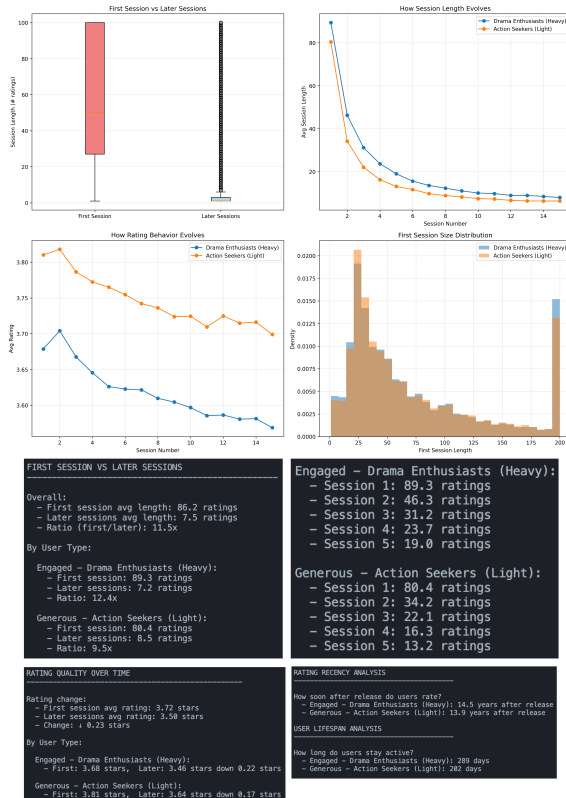


Fig. 6. User Lifecycle (Temporal Patterns)

	Engaged Raters	Generous Raters
First Session Ratings	89.3	80.4
Later Session Ratings	7.2	8.5
Burst Ratio (first/later)	12.4x	9.5x
Rating Recency (years after release)	14.5	13.9
Genre Diversity Score	16.49	16.38
Rating Decline (first to later)	3.68 to 3.46 (-0.22)	3.81 to 3.64 (-0.17)
Lifespan (days)	289	202

Users rate far more movies in their first session than later ones, indicating a strong “onboarding” effect. This decline is steeper for Engaged Raters, reflecting higher initial activity. Engaged Raters also explore slightly older movies, suggesting deeper catalog engagement, while Generous Raters lean more toward newer releases. Despite this, both groups exhibit similar genre diversity, indicating equally broad tastes but different preferences. In terms of retention, Engaged Raters remain active significantly longer, reinforcing their higher overall engagement. Finally, both groups give lower ratings over time, suggesting initial ratings are inflated by early enthusiasm. Generous Raters consistently give higher ratings and show a smaller decline, indicating a more stable and lenient rating pattern.

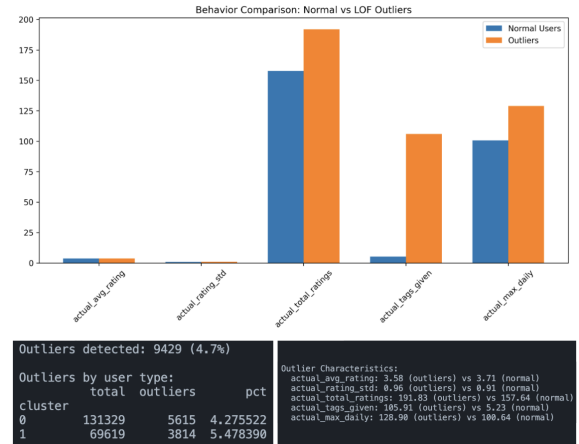


Fig. 7. Normal User Behaviors VS. Outliers

The LOF applied using behavioral features (average rating, rating consistency, rating volume, tagging activity, and maximum daily ratings) for anomaly detection identified 4.7% of users as behavioral outliers based on the natural distribution of outlier scores. Generous Raters showed a higher outlier rate compared to Engaged Raters. The outlier users were extreme across multiple engagement dimensions as they had 20 times more tagging activity, higher rating volume, greater rating inconsistency, and more binge behavior with higher maximum daily ratings.

D. Cross-Domain Patterns

The cross-domain analysis examining how user types rate different movie clusters reveals that Generous Raters rate all movie categories higher than Engaged Raters (0.04–0.08 stars

higher, hence why the name was given). Furthermore, both user types agree on the hierarchy of movie clusters, with Classics receiving the highest ratings from both groups.

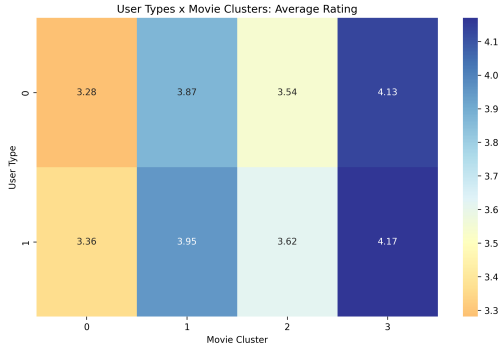


Fig. 8. User Types vs. Movie Clusters Heatmap

V. DISCOVERY QUESTIONS

1. What natural groups of movies exist based on user ratings, genres, and tags?

Based on the clustering and validation analysis, four natural groups of movies exist that form a hierarchy of audience engagement. The Long Tail, comprising 93.8% of movies, represents obscure films with low engagement, only 426 ratings per movie and average ratings of 3.14 stars. The Mid-Tier Successes, 5.3% of movies, achieve moderate engagement with 6,915 ratings and 3.47 stars average ratings. Popular Hits, only 0.8% of movies, achieve high engagement with 23,742 ratings but only 3.83 star average ratings, demonstrating that popularity does not guarantee quality. All-Time Classics, the rarest group at 0.07% of movies, achieve both massive popularity with 56,145 ratings and exceptional quality with 4.13 star average ratings.

The decision tree validation confirms that these groups are primarily driven by popularity (rating_count at 80.9% importance) rather than quality (avg_rating at 4.4% importance), meaning that audience engagement defines movie tiers more than critical acclaim. Genre analysis reveals that Action, Sci-Fi, and Adventure dominate Popular Hits, while Drama and Thriller define Classics. This answers the question by showing that natural movie groups emerge from the interplay between popularity and quality, with most films occupying the long tail of low engagement and only a tiny fraction achieving classic status through both mass appeal and critical excellence.

2. How do user engagement patterns (rating volume and tagging) relate to rating behavior, and how do these factors form distinct user archetypes?

The analysis reveals two distinct user archetypes defined by different engagement profiles. Engaged Raters, comprising 65.4% of users, demonstrate higher rating volume (175 vs 129

ratings) and tagging activity (12.1 vs 5.8 tags), but give lower average ratings (3.66 vs 3.79 stars). They are predominantly Drama enthusiasts. Generous Raters, comprising 34.6% of users, rate fewer movies and tag less, but give consistently higher ratings across all categories and show more consistent rating patterns. They are predominantly Action and Adventure enthusiasts.

The decision tree validation using only behavioral features shows that rating quality (avg_rating: 51.7%) and rating consistency (rating_std: 21.5%) are the strongest differentiators between archetypes, more important than rating volume (16.6%) or tagging behavior (9.7%). This indicates that how users rate matters more than how much they rate or what they watch.

The relationship between engagement patterns and rating behavior is therefore that users separate into two archetypes based primarily on rating quality and consistency. Engaged Raters prioritize volume and depth of engagement (more ratings, more tags, longer lifespan) but are slightly more critical. Generous Raters prioritize selectivity and generosity (higher ratings, more consistency) but engage with fewer movies. Genre preferences correlate with these behaviors (Drama with volume-driven engagement, Action with quality-driven engagement) but are not the primary differentiator.

3. How do temporal patterns in rating activity (burst behavior, rating recency, and consistency over time) differ between user archetypes, and what anomalous rating patterns emerge?

Temporal patterns differ significantly between the two user archetypes. Regarding burst behavior, first sessions are dramatically larger than later sessions for both groups, with users rating 11.5 times more movies in their initial session. Engaged Raters show a steeper decline (12.4x ratio, 89.3 to 7.2 ratings) compared to Generous Raters (9.5x ratio, 80.4 to 8.5 ratings), reflecting their higher initial engagement. This indicates universal onboarding behavior where users explore actively before settling into routine, with Engaged Raters showing more extreme initial engagement.

Regarding rating recency, Engaged Raters rate movies an average of 14.5 years after release, while Generous Raters rate movies 13.9 years after release. This 0.6-year difference indicates that Engaged Raters explore deeper into the catalog, watching older films, while Generous Raters focus somewhat more on new releases. Notably, both groups have nearly identical genre diversity scores (16.49 vs 16.38), meaning their taste breadth is similar, they simply prefer different genres.

Regarding consistency over time, both user types show decreasing ratings from first to later sessions. Engaged Raters decline from 3.68 to 3.46 stars (-0.22), while Generous Raters decline from 3.81 to 3.64 stars (-0.17). This suggests first-session rating inflation is universal showing that users rate more generously when first joining. Generous Raters maintain higher ratings throughout and show a smaller decline, confirming they are consistently more generous raters.

Regarding lifespan, Engaged Raters remain engaged for 289 days on average compared to 202 days for Generous Raters, an 87-day difference confirming that the more volume-engaged archetype also sustains engagement much longer.

For anomalous rating patterns, Local Outlier Factor analysis with automatic contamination detection identified 4.7% of users as behavioral outliers. These users were characterized by extreme tagging behavior (105.9 vs 5.2 tags for normal users; 20 times more), higher rating volume (191.8 vs 157.6 ratings), and greater inconsistency in ratings (std deviation 0.96 vs 0.91). Generous Raters showed a higher outlier rate (5.5%) than Engaged Raters (4.3%), suggesting that the more selective, quality-driven archetype exhibits more behavioral extremes. These anomalies represent true power users at the extreme end of engagement who rate extensively, tag prolifically, and hold more variable opinions.

VI. M4 PLANS

For M4, the focus will be on consolidating the analytical pipeline into a polished, reproducible, and well-documented final study. Additional validation will be performed on clustering stability (varying K values, random seeds, and feature subsets) to confirm robustness of both movie groups and user archetypes. Visualizations will be refined for clarity, including improved PCA cluster plots, feature importance charts, and temporal behavior graphs. The report will integrate all discovery question findings into a cohesive narrative supported by figures and tables, business insights, and the README will be expanded so the entire project can be executed end-to-end by cloning the repository and running the scripts in sequence. Final emphasis will be placed on clearly connecting analytical techniques (PCA, K-Means, Decision Trees, and LOF) to the patterns they revealed and the discovery questions they answered.

VII. CONCLUSION

This project demonstrates that meaningful structure exists in both movie and user behavior data when examined through clustering, dimensionality reduction, classification, and anomaly detection. Movies naturally organize into a hierarchy defined primarily by audience engagement rather than rating quality, revealing a pronounced long tail and a tiny set of true classics. Users separate into two archetypes driven more by rating behavior and consistency than by genre preference, showing that how users rate is more important than what they watch. Temporal analysis further reveals universal onboarding bursts, rating inflation in early sessions, and lifespan differences between archetypes. Finally, anomaly detection shows that while most users fit these patterns, a small fringe exhibits behaviors beyond the primary groups. Together, these findings provide a comprehensive answer to the discovery questions and establish a strong analytical foundation for the final deliverable.

REFERENCES

- [1] "Movielens," GroupLens, <https://grouplens.org/datasets/movielens/>.